

Daily Geometry Drill #1 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Gradients & Perpendicularity · Distance & Midpoints · Perpendicular Distance · Shoelace Areas · Collinearity.

Section A Rapid Recognition

1. Find the gradient of the line $6x - 3y + 15 = 0$.

Answer: 2

Method: Rearrange to gradient–intercept form: $3y = 6x + 15 \Rightarrow y = 2x + 5$, so $m = 2$.

Investigate further: Implicit form $ax + by + c = 0$ has gradient $-a/b$; here $-\frac{6}{3} = 2$. **If the line is instead written $3y - 6x - 15 = 0$ (the same line), is the gradient still 2?** It must be — check the $-a/b$ rule still agrees.

2. Find the gradient of the line through $(1, 2)$ and $(5, 14)$.

Answer: 3

Method: $m = \frac{14-2}{5-1} = \frac{12}{4} = 3$.

Investigate further: Check with $(5 - 1, 14 - 2)$ swapped: $\frac{2-14}{1-5} = \frac{-12}{-4} = 3$ ✓ — the sign always cancels. **If the line through $(1, 2)$ and $(5, 14)$ is moved up by 3 (every y increased by 3), does its gradient change?** No — a vertical translation keeps the ratio of rises fixed.

3. A line of gradient 4 is perpendicular to a line of gradient m . Write down m .

Answer: $-\frac{1}{4}$

Method: Perpendicular gradients are negative reciprocals: $m = -\frac{1}{4}$, since $4 \cdot (-\frac{1}{4}) = -1$.

Investigate further: The product test $m_1 m_2 = -1$ is the fastest check. **Is the line $y = 4x$ parallel, perpendicular, or neither, to $y = -\frac{1}{4}x + 12$?** The constant term is a distraction — gradients alone decide.

4. Find the distance between $(2, 1)$ and $(5, 5)$.

Answer: 5

Method: $d = \sqrt{(5-2)^2 + (5-1)^2} = \sqrt{9+16} = \sqrt{25} = 5$.

Investigate further: 3-4-5 right triangle in disguise. **The points $(-1, 0)$ and $(3, -3)$ are the same distance apart — can you see the Pythagorean triple before computing, by drawing the displacement vector?** The displacement $(4, -3)$ is a reordered 3-4-5.

5. Find the midpoint of the segment joining $(4, 8)$ and $(10, 2)$.

Answer: $(7, 5)$

Method: Midpoint averages each coordinate: $\left(\frac{4+10}{2}, \frac{8+2}{2}\right) = (7, 5)$.

Investigate further: Check both halves are equal distances: $(7-4, 8-5) = (3, 3)$ and $(10-7, 5-2) = (3, 3) \checkmark$. **The midpoint of two points is the average of the two displacement vectors. What is the midpoint of $(4, 8)$ and $(-6, -2)$?** Negative coordinates behave the same way.

6. Find the perpendicular distance from the origin to the line $3x + 4y - 10 = 0$.

Answer: 2

Method: Use $\frac{|ax_0+by_0+c|}{\sqrt{a^2+b^2}}$ at $(0, 0)$: $\frac{|-10|}{\sqrt{9+16}} = \frac{10}{5} = 2$.

Investigate further: The denominator 5 is the length of the normal vector $(3, 4)$. **The point of the line closest to the origin satisfies $(x, y) = t(3, 4)$ for some t — find it and confirm its distance is indeed 2.** Every point of the line's coordinate vector is a multiple of its normal.

7. A line has x -intercept 4 and y -intercept 6. Find its gradient.

Answer: $-\frac{3}{2}$

Method: The intercepts are the points $(4, 0)$ and $(0, 6)$: $m = \frac{6-0}{0-4} = -\frac{3}{2}$.

Investigate further: Reading right-to-left: this is the negative of $\frac{y\text{-intercept}}{x\text{-intercept}}$. **Write the line in intercept form $\frac{x}{4} + \frac{y}{6} = 1$ and rearrange to confirm $m = -\frac{y\text{-intercept}}{x\text{-intercept}} = -\frac{6}{4} = -\frac{3}{2}$ — the intercept form makes the gradient transparent.**

8. Find the area of the triangle with vertices $(0, 0)$, $(4, 0)$ and $(0, 3)$.

Answer: 6

Method: Right-angled at the origin: area = $\frac{1}{2} \cdot 4 \cdot 3 = 6$.

Investigate further: Equivalently the shoelace/midpoint-vector formula $\frac{1}{2}|(4)(3) - (0)(0)| = 6$. **Rotate this triangle by 90° about the origin: does the area change, and do you get the same shoelace number?** Reflections and rotations preserve area.

9. The points $(1, 2)$, $(2, 4)$ and $(k, 10)$ are collinear. Find k .

Answer: 5

Method: Rise per unit run is 2 (from $(1, 2)$ to $(2, 4)$); continuing to $y = 10$ climbs 6 more, so $k = 2 + \frac{6}{2} = 5$.

Investigate further: Formal check: $\frac{4-2}{2-1} = \frac{10-4}{k-2} \Rightarrow 2 = \frac{6}{k-2} \Rightarrow k = 5$. **What happens to k if the points are required to be collinear but the order is $(2, 4)$, $(1, 2)$, $(k, 10)$?** Collinearity is order-independent — all three gradients must agree.

10. Using the shoelace formula, find the area of the triangle with vertices $(0, 0)$, $(3, 4)$ and $(6, 0)$.

Answer: 12

Method: Shoelace: $\frac{1}{2}|0 \cdot 4 + 3 \cdot 0 + 6 \cdot 0 - (0 \cdot 3 + 4 \cdot 6 + 0 \cdot 0)| = \frac{1}{2}|24| = 12$.

Investigate further: Bonus route: base = 6, height = 4, area = $\frac{1}{2} \cdot 6 \cdot 4 = 12$ ✓. **For a triangle with one vertex at the origin, shoelace collapses to $\frac{1}{2}|x_1y_2 - x_2y_1|$ with the other two points — prove that this equals $\frac{1}{2}|\det|$ of the two vectors.** That determinant is the signed area.

Section B Manipulation Drills

1. The line $2x - 3y + 6 = 0$ is perpendicular to which of the following?

- A) $y = -\frac{3}{2}x + 1$
- B) $y = \frac{3}{2}x - 2$
- C) $y = \frac{2}{3}x + 5$
- D) $y = -\frac{2}{3}x + 3$

Answer: A) $y = -\frac{3}{2}x + 1$

Method: Given line has gradient $\frac{2}{3}$ (rule $-a/b$), so perpendicular gradient is $-\frac{3}{2}$. Only option A has gradient $-\frac{3}{2}$.

Investigate further: Options B and C have the same gradient, and D the new “same slope” decoy — read *gradients only* through the $-\frac{3}{2}$ filter. **Which option is parallel to the given line, and which is merely reflection-symmetric in slope?** Perpendicularity and parallelism are decided by one number, the slope.

2. What is the perpendicular distance from $(2, 3)$ to the line $5x + 12y - 7 = 0$?

- A) 1
- B) 2
- C) 3
- D) 4

Answer: C) 3

Method: $\frac{|5(2)+12(3)-7|}{\sqrt{25+144}} = \frac{|10+36-7|}{13} = \frac{39}{13} = 3$.

Investigate further: The numerator is exactly how far the point sits from the line measured along a normal whose length is 13. **Two points are at perpendicular distance 3 from $5x + 12y - 7 = 0$ — what do they have in common?** They lie on $5x + 12y - 7 = \pm 39$, the two parallel satellite lines.

3. Which point lies on the perpendicular bisector of the segment joining $(1, 2)$ and $(5, 6)$?

- A) $(2, 5)$
- B) $(3, 3)$
- C) $(4, 4)$
- D) $(5, 3)$

Answer: A) (2, 5)

Method: Midpoint (3, 4); segment gradient 1, so bisector gradient -1 : $y - 4 = -(x - 3)$, i.e. $x + y = 7$. Check options: $2 + 5 = 7$ ✓.

Investigate further: Only A satisfies $x + y = 7$. **A point is on the perpendicular bisector iff it is equidistant from the two endpoints — verify (2, 5) is distance $\sqrt{16}$ from both (1, 2) and (5, 6).** That's the defining property, worth internalising.

4. Find the area of the quadrilateral with vertices (0, 0), (4, 0), (5, 3), (1, 3), in that order.

- A) 10
B) 12
C) 14
D) 16

Answer: B) 12

Method: Shoelace in order: $\frac{1}{2}|0 \cdot 0 + 4 \cdot 3 + 5 \cdot 3 + 1 \cdot 0 - (0 \cdot 4 + 0 \cdot 5 + 3 \cdot 1 + 3 \cdot 0)| = \frac{1}{2}|0 + 12 + 15 + 0 - (0 + 0 + 3 + 0)| = \frac{1}{2}|24| = 12$.

Investigate further: The quadrilateral is a parallelogram (bases $\parallel$ via position vectors (4, 0) and (1, 3)), so area = $|\det| = |4(3) - 0(1)| = 12$ ✓. **Reorder the vertices as (0, 0), (5, 3), (4, 0), (1, 3): the same region, but the shoelace number warns you when points are listed in the wrong order.** Keep vertices in cyclic order.

5. The line through (1, 2) and (4, 5) crosses the y -axis at which value?

- A) 0
B) 1
C) 2
D) 3

Answer: B) 1

Method: Gradient $\frac{5-2}{4-1} = 1$; $y = x + c$ through (1, 2) gives $c = 1$.

Investigate further: Extension: at $x = 0$, $y = 1$. **Through the point-slope form, $y - 2 = 1(x - 1)$. If the line were instead the one through (1, 2) and (4, 8), the y -intercept changes — resolve c again.** Always plug the known point in, never guess c from the picture.

6. Find the perpendicular distance from (0, 1) to the line $6x - 8y + 2 = 0$.

Answer: $\frac{3}{5}$

Method: $\frac{|6(0) - 8(1) + 2|}{\sqrt{36 + 64}} = \frac{|-6|}{10} = \frac{3}{5}$.

Investigate further: Normal length $\sqrt{36 + 64} = 10$; numerator 6. **The line $6x - 8y + 2 = 0$ sits distance $\frac{1}{5}$ from the origin (check $\frac{|2|}{10}$). Since the y -axis point (0, 1) is 1 unit up from the origin, the net distance $|1 - \frac{1}{5}| = \frac{4}{5}$ differs from $\frac{3}{5}$ — why?** Because the normal (6, -8) is not vertical: only the projection along it counts.

7. Which of the following is NOT parallel to $y = 3x + 1$?

- A) $y = 3x + 5$
- B) $6x - 2y = 4$
- C) $3x - y = 2$
- D) $y = 2x + 1$

Answer: D) $y = 2x + 1$

Method: Parallel $\Leftrightarrow$ same gradient 3. A: $3\checkmark$; B: $6x - 2y = 4 \Rightarrow y = 3x - 2\checkmark$; C: $3x - y = 2 \Rightarrow y = 3x - 2\checkmark$; D: $y = 2x + 1$ has gradient 2.

Investigate further: Options B and C are disguises (divide by the y -coefficient first). **Show two distinct parallel lines never intersect, while any two non-parallel lines always do.** Parallelism is exactly “same slope, different intercept”.

8. The line $y = x + 1$ meets the line through $(0, 3)$ perpendicular to it at the point P . Find the coordinates of P .

Answer: $(1, 2)$

Method: Through $(0, 3)$ with gradient -1 : $y = -x + 3$. Intersect: $x + 1 = -x + 3 \Rightarrow x = 1$, then $y = 2$.

Investigate further: The intersection of two perpendicular lines is the right-angle vertex of a right triangle. Verify $P = (1, 2)$ lies on both lines: $y = x + 1$ gives $2 = 1 + 1 \checkmark$ and $y = -x + 3$ gives $2 = -1 + 3 \checkmark$. **Through $(0, 3)$ draw the perpendicular and reflect the point where it meets $y = x + 1$ across that meeting point — what do you get, and how does it connect to the midpoint of $(0, 3)$ and $(2, 1)$?** Perpendicular lines meet exactly once; the reflect-and-extend trick answers the follow-up.

9. Which point is closest to the origin?

- A) $(1, 3)$
- B) $(2, 2)$
- C) $(4, 0)$
- D) $(0, 5)$

Answer: B) $(2, 2)$

Method: Distances: $\sqrt{(1, 3)} = \sqrt{10}$, $\sqrt{(2, 2)} = \sqrt{8}$, $\sqrt{(4, 0)} = 4$, $\sqrt{(0, 5)} = 5$; smallest is $\sqrt{8}$.

Investigate further: Comparing squared distances avoids surds: 10, 8, 16, 25. **Among points with x, y nonnegative integers and $x + y = 4$, which is closest to the origin?** Both squares and sum compare against $(2, 2)$.

10. Using the shoelace formula, find the area of the triangle with vertices $(1, 1)$, $(4, 2)$, $(2, 6)$.

Answer: 7

Method: $\frac{1}{2}|1 \cdot 2 + 4 \cdot 6 + 2 \cdot 1 - (1 \cdot 4 + 2 \cdot 2 + 6 \cdot 1)| = \frac{1}{2}|2 + 24 + 2 - (4 + 4 + 6)| = \frac{1}{2}|28 - 14| = 7$.

Investigate further: The 7 here is exactly the determinant area of the two vectors $(3, 1)$ and

$(1, 5): \frac{1}{2}|3(5) - 1(1)| = \frac{1}{2}(14) = 7 \checkmark$. **In shoelace, matching – and cancelling – the repeated coordinate arithmetic is a good fastest-check habit.** Confirm you did $\frac{1}{2}|(x_1y_2 + x_2y_3 + x_3y_1) - (y_1x_2 + y_2x_3 + y_3x_1)|$.

Section C Substitution & Structure

1. The midpoint of $(2, -1)$ and $(6, 7)$ is?

- A) $(4, 3)$
- B) $(3, 4)$
- C) $(5, 4)$
- D) $(4, 2)$

Answer: A) $(4, 3)$

Method: $\left(\frac{2+6}{2}, \frac{-1+7}{2}\right) = (4, 3)$.

Investigate further: A and B swap the coordinates of the answer; D averages the x 's but not the y 's. **Midpoint of $(2, -1)$ and $(6, 7)$ is $(4, 3)$; of $(2, -1)$ and $(6, 7)$ reflected in the origin, $(-6, -7)$, is $(-2, -4)$.** Averaging negatives works the same.

2. The perpendicular bisector of the segment joining $(0, 1)$ and $(4, 5)$ has equation?

- A) $x + y = 5$
- B) $y = x + 1$
- C) $y = -x + 3$
- D) $y = x + 3$

Answer: A) $x + y = 5$

Method: Midpoint $(2, 3)$; segment gradient $\frac{5-1}{4-0} = 1$, so perpendicular gradient -1 : $y - 3 = -(x - 2) \Rightarrow x + y = 5$.

Investigate further: B and D are the perpendicular-doubles (gradient 1); C is a parallel shifted copy missing the midpoint. **The perpendicular bisector is the locus of equidistant points — verify $(2, 3)$ and, say, $(6, -1)$ both satisfy $x + y = 5$ and check equal distances to both endpoints.**

3. The distance between $(2, 5)$ and $(7, 17)$ is?

- A) 13
- B) 14
- C) $\sqrt{194}$
- D) 12

Answer: A) 13

Method: $\sqrt{(7-2)^2 + (17-5)^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.

Investigate further: 5-12-13 triple again; option C is the “forgot to take the square root” trap ($25 + 144 = 169$, not $\sqrt{194}$). **If the second point were (7, 13) instead, the distance $\sqrt{25 + 64} = \sqrt{89}$ stays a surd — prove it is irrational.** Rational squares never land on 89.

4. The area of the triangle with vertices $(0, 0)$, $(5, 0)$, $(2, 6)$ is?

- A) 15
- B) 30
- C) 10
- D) 12

Answer: A) 15

Method: Take $(5, 0)$ as base, height 6 from $(2, 6)$: $\frac{1}{2} \cdot 5 \cdot 6 = 15$.

Investigate further: Shoelace check: $\frac{1}{2}|0 + 5(6) + 0 - (0 + 0 + 0)| = 15 \checkmark$. **The altitude is measured to the base line $y = 0$ — the 6 is the y -coordinate. If the apex were $(2, -6)$ the area is still 15; is that a reflection or a genuine second triangle?** Area is absolute.

5. Which line is perpendicular to $y = 2x - 3$?

- A) $y = -2x + 1$
- B) $y = -\frac{1}{2}x + 4$
- C) $y = \frac{1}{2}x$
- D) $y = 2x + 5$

Answer: B) $y = -\frac{1}{2}x + 4$

Method: Gradient 2 flips and inverts to $-\frac{1}{2}$.

Investigate further: A is the “negate only” trap, C the “invert only” trap, D parallel. **Which lines through $(0, 4)$ are perpendicular to $y = 2x - 3$, and how many are there?** Through any fixed point there is exactly one perpendicular line: $y = -\frac{1}{2}x + 4$ here.

6. The points $(1, 4)$, $(4, 7)$, $(10, k)$ are collinear. Find k .

- A) 10
- B) 12
- C) 13
- D) 15

Answer: C) 13

Method: Gradient between the first two is $\frac{7-4}{4-1} = 1$; at $x = 10$, six units further, $y = 4 + \frac{10-1}{1} \cdot 1 = 13$.

Investigate further: Equivalently $k = 4 + (10 - 1) = 13$, and $\frac{k-7}{10-4} = 1 \Rightarrow k = 13 \checkmark$. **If the middle point were removed, the “two” gradients that must agree are $(1, 4) \rightarrow (10, k)$ and $(1, 4) \rightarrow (4, 7)$ — verify all three pairwise gradients equal 1.**

7. The perpendicular distance from $(2, -1)$ to the line $3x - 4y + 10 = 0$ is?

- A) 2
- B) 3
- C) 4
- D) 5

Answer: C) 4

Method: $\frac{|3(2) - 4(-1) + 10|}{\sqrt{9+16}} = \frac{|6+4+10|}{5} = \frac{20}{5} = 4$.

Investigate further: Big numerator: the point lies far out along the normal. **Recompute with the point $(2, 1)$:** $\frac{|6-4+10|}{5} = \frac{12}{5} = 2.4$; **the distance changes as the point slides, while the normal length 5 stays fixed.** The numerator measures signed offset along the normal.

8. The line through $(2, 1)$ perpendicular to $y = \frac{1}{2}x + 3$ has equation?

- A) $y = -2x + 5$
- B) $y = 2x - 3$
- C) $y = -\frac{1}{2}x + 1$
- D) $y = -2x + 3$

Answer: A) $y = -2x + 5$

Method: Gradient $-\frac{1}{1/2} = -2$; through $(2, 1)$: $y - 1 = -2(x - 2) \Rightarrow y = -2x + 5$.

Investigate further: Options A and D share gradient -2 but only A passes through $(2, 1)$ (check $1 = -4 + 5$ ✓). **The line $y = -2x + 5$ and $y = \frac{1}{2}x + 3$ meet at $x = 0.8, y = 3.4$ — verify the product of gradients $(-2)(\frac{1}{2}) = -1$ ✓.** Perpendicular lines always cross at right angles, wherever that is.

Section D Challenge

1. The perpendicular bisector of the segment joining $(3, -1)$ and $(7, 5)$ meets the x -axis at P . What is the x -coordinate of P ?

- A) 6
- B) 7
- C) 8
- D) 9

Answer: C) 8

Method: Midpoint $(5, 2)$; segment gradient $\frac{5-(-1)}{7-3} = \frac{3}{2}$, so bisector gradient $-\frac{2}{3}$: $y - 2 = -\frac{2}{3}(x - 5)$. At $y = 0$: $-2 = -\frac{2}{3}(x - 5) \Rightarrow x - 5 = 3 \Rightarrow x = 8$.

Investigate further: General trick: the perpendicular bisector of a segment is equidistant from both ends, so $P = (p, 0)$ satisfies $\sqrt{(p-3)^2 + 1} = \sqrt{(p-7)^2 + 25}$. Squaring: $(p-3)^2 + 1 = (p-7)^2 + 25 \Rightarrow p = 8$ ✓. **Both routes agree — the equidistance route never needs a slope.**

2. A triangle with vertices $(0, 0)$, $(6, 0)$ and $(3, h)$ has area 12. Find h .

- A) 2
B) 4
C) 6
D) 8

Answer: B) 4

Method: Base 6 on the x -axis; altitude is $|h|$: $\frac{1}{2} \cdot 6 \cdot |h| = 12 \Rightarrow h = \pm 4$.

Investigate further: Options B and D double the right answer? No — B is 4, D is 8; both signs are valid, the question intends the positive one. **The apex $(3, h)$ is directly above the midpoint of the base, making an isosceles triangle. What is its perimeter at $h = 4$?** Sides are $x = 3, y = 4$ hypotenuses of length 5: perimeter = $6 + 10 = 16$.

3. The distance from the point $(4, 3)$ to the line $y = x$ is?

- A) $\frac{1}{\sqrt{2}}$
B) 1
C) $\sqrt{2}$
D) $\frac{7}{\sqrt{2}}$

Answer: A) $\frac{1}{\sqrt{2}}$

Method: Line $x - y = 0$: $\frac{|4-3|}{\sqrt{1^2+(-1)^2}} = \frac{1}{\sqrt{2}}$.

Investigate further: For $y = x$ the distance formula halves to $|x_0 - y_0|/\sqrt{2}$ — the 1 difference over $\sqrt{2}$. **The point $(4, 3)$ lies $\frac{1}{\sqrt{2}}$ from $y = x$ and $\frac{7}{\sqrt{2}}$ from $y = -x$ (check $|4 + 3|/\sqrt{2}$). What is the distance between the two lines $y = x$ and $y = -x$ themselves?** It's 0 — they meet at the origin; only non-parallel pairs have a “distance” in this sense, and these two are perpendicular.

4. The line $2x + 3y - 12 = 0$ cuts the coordinate axes to form a right-angled triangle. Its area is?

- A) 6
B) 8
C) 12
D) 24

Answer: C) 12

Method: Intercepts: $x = 6$ at $y = 0$, $y = 4$ at $x = 0$, so area = $\frac{1}{2} \cdot 6 \cdot 4 = 12$.

Investigate further: The intercept form $\frac{x}{6} + \frac{y}{4} = 1$ reads intercepts straight off. **If instead the line were $4x + 3y - 12 = 0$, the intercepts are 3 and 4 — which axis-touching area, 6 or 12, does the swap produce?** Check $\frac{1}{2} \cdot 3 \cdot 4 = 6$; the 12 in the defining equation is *not* the area.

5. Using the shoelace formula, find the area of the quadrilateral with vertices $(0, 2)$, $(3, 0)$, $(6, 4)$, $(1, 6)$, in that order.

Answer: 20

Method: Shoelace in order: $\frac{1}{2} |(0)(0) + (3)(4) + (6)(6) + (1)(2) - [(2)(3) + (0)(6) + (4)(1) + (6)(0)]| = \frac{1}{2} |(0 + 12 + 36 + 2) - (6 + 0 + 4 + 0)| = \frac{1}{2} |50 - 10| = 20$.

Investigate further: Cross-check by triangulating along $(0, 2) \rightarrow (6, 4)$: the two triangles have areas 9 and 11, summing to 20 ✓. **Now list the same four points in the order** $(0, 2)$, $(3, 0)$, $(1, 6)$, $(6, 4)$. **Compute the shoelace value — it differs, because that polygon crosses itself.** Reconcile the two, and note that area formulas assume a simple (non-crossing) polygon.

Top 5 Patterns — Worksheet #1

- 1. Implicit line $\rightarrow$ gradient: rearrange to $y = mx + c$ (or read $-a/b$).** The constant c never touches m — strip it off last. (see A1, A7, B5)
- 2. Perpendicular gradients are negative reciprocals:** $m_1 m_2 = -1$. Slack jaw never flips and negates; keep both operations. (see A3, B1, B8, C5, C8)
- 3. Perpendicular bisector = midpoint + negative-reciprocal slope.** Always hit the midpoint first; the bisector is the line through it with the flipped-inverted gradient. (see B3, C2, D1)
- 4. Perpendicular distance formula:** $\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$. Numerator from substitution, denominator from the normal vector's length. (see A6, B2, B6, C7, D3)
- 5. Shoelace (signed) area:** $\frac{1}{2} |\sum x_i y_{i+1} - \sum y_i x_{i+1}|$. Vertices in cyclic order; the $\frac{1}{2}$ and the absolute value are non-negotiable. (see A10, B4, B10, D5)

Common Traps — Worksheet #1

- 1. Sign of the implicit gradient.** $6x - 3y + 15 = 0$ has gradient 2, not -2 and not 3 — always solve for y explicitly, or read $-a/b$ with both signs. (see A1, A7, B1)
- 2. Distance: absolute value & square root together.** $\frac{|\dots+c|}{\sqrt{\dots}}$ — dropping the absolute value floors the answer at zero and loses the sign of the offset. (see A6, B2, C7)
- 3. Perpendicular does not mean “negated slope”.** $m = 2$ pairs with $-\frac{1}{2}$, not -2 ; only the flip-invert passes $m_1 m_2 = -1$. (see C5, C8, B1)
- 4. Shoelace: half the absolute difference, vertices in order.** Out-of-order vertices produce self-crossing regions whose signed sum misleads — keep cyclic order and take half of the magnitude. (see B4, B10, D5)
- 5. Collinearity: compare the right gradient pairs.** Three points are collinear iff *every* pair gives the same slope — one matching pair is never enough. (see A9, C6)

“Shoelace before altitude: area is a one-line cross product, not a construction.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths